

PROJECT REPORT

SSH Brute Force Attack Detection Using Splunk SIEM

Prepared by

Avishkar Salve

SOC Analyst Fresher | Splunk | Incident Monitoring | SIEM
Cybersecurity | Certified Ethical Hacker (CEH)

Security Operations Center (SOC) Use Case | Splunk Enterprise | Linux Authentication Logs

Abstract

Cyber attacks targeting authentication services are among the most common threats faced by organizations. Brute force attacks attempt to gain unauthorized access by repeatedly trying different username and password combinations. This project demonstrates how Splunk SIEM can be used to detect SSH brute force attacks by analyzing Linux authentication logs, identifying suspicious login patterns, and generating security dashboards for monitoring and investigation.

The project simulates a real-world Security Operations Center (SOC) use case where analysts monitor authentication events, investigate suspicious activity, and respond to potential compromises.

1. Introduction

Organizations rely on Secure Shell (SSH) for remote administration of Linux systems. Attackers frequently target SSH services by performing brute force attacks to obtain valid credentials.

Security Information and Event Management (SIEM) platforms such as Splunk help security teams detect these attacks by collecting, analyzing, and correlating log data from multiple sources.

This project focuses on detecting SSH brute force attacks using Linux authentication logs and Splunk SPL queries.

2. Objectives

The main objectives of this project are:

- Collect Linux authentication logs
- Detect repeated failed login attempts
- Identify successful logins following multiple failures
- Monitor attacker IP addresses
- Analyze targeted user accounts
- Create dashboards for SOC monitoring
- Improve threat detection and incident response capabilities

3. Problem Statement

Brute force attacks generate a large number of authentication events that may go unnoticed without proper monitoring.

Security analysts require a solution capable of:

- Identifying suspicious authentication behavior
- Detecting account compromise attempts
- Visualizing attack trends

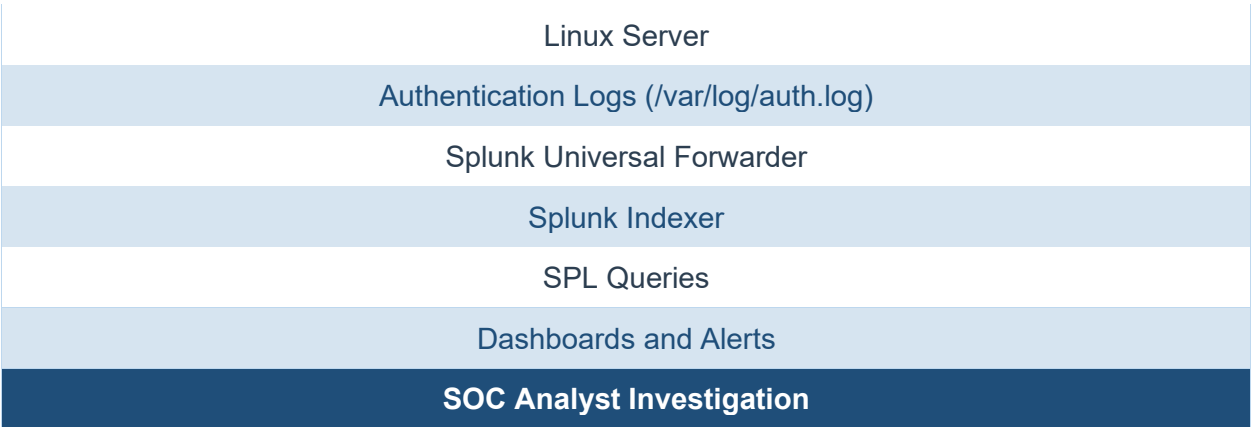
- Supporting incident investigation

4. Tools and Technologies

Tool / Technology	Purpose
Splunk Enterprise	Log Analysis and SIEM
Splunk Universal Forwarder	Log Collection
Linux Authentication Logs	Data Source
SPL (Search Processing Language)	Threat Detection
Ubuntu Linux	Log Generation
Windows	Splunk Server Host

5. System Architecture

The following pipeline describes the data flow from log generation to analyst investigation:



6. Data Source

The project uses Linux authentication logs generated from SSH login activities.

Log Location

```
/var/log/auth.log
```

Events Monitored

- Failed password
- Accepted password
- Invalid user
- Authentication failure

7. Detection Methodology

The detection process follows these sequential steps:

1. Collect Linux authentication logs
2. Forward logs to Splunk
3. Index and normalize log events
4. Analyze failed login attempts
5. Correlate successful logins after failures
6. Generate dashboards and alerts

8. SPL Query for Brute Force Detection

```
index=linux_log sourcetype=linux_secure_log ("Failed password" OR "Accepted password")
| eval action=if(searchmatch("Failed password"),"Failed","Success")
| stats count(eval(action="Failed")) as Failed_Attempts,
          count(eval(action="Success")) as Successful_Logins
  by src_ip, host_name
| where Failed_Attempts>=5 AND Successful_Logins>=1
```

Query Explanation

- Detects both failed and successful SSH logins
- Counts failed login attempts per source IP
- Counts successful logins per source IP
- Groups events by source IP address and hostname

The query flags suspicious activity when the following conditions are both met:

- Failed attempts ≥ 5
- Successful login ≥ 1

Note: This pattern — multiple failures followed by a success — may indicate successful credential compromise and should be investigated immediately.

9. Dashboard Components

9.1 Failed SSH Login Trend

Purpose:

- Monitor failed authentication attempts over time

Benefits:

- Detect attack spikes
- Identify unusual login activity

9.2 Top Attacker IPs

Purpose:

- Identify IP addresses generating the highest number of failed logins

Benefits:

- Detect malicious source addresses
- Support IP blocking decisions

9.3 Top Targeted Users

Purpose:

- Identify accounts targeted during brute force attacks

Benefits:

- Detect high-risk accounts
- Improve account protection measures

9.4 Brute Force Followed by Success

Purpose:

- Detect successful logins after repeated failures

Benefits:

- Identify possible account compromise
- Prioritize incident investigation and response

10. MITRE ATT&CK Mapping

Technique ID	Technique Name
T1110	Brute Force
T1078	Valid Accounts

11. SOC Investigation Workflow

Upon receiving a Splunk alert, the SOC analyst follows this investigation workflow:

7. Receive alert from Splunk
8. Review attacker IP address
9. Analyze targeted user account
10. Verify successful login events
11. Check host activity for anomalies
12. Determine compromise impact
13. Escalate incident if necessary
14. Recommend and implement remediation

12. Results

The project successfully achieved the following outcomes:

- Detected brute force attempts in real time
- Identified suspicious source IP addresses
- Highlighted targeted user accounts
- Detected successful authentication events following repeated failures
- Provided visual dashboards for ongoing SOC monitoring

13. Challenges Faced

- Log parsing and field extraction from raw auth.log
- Event correlation across multiple log entries
- Identifying meaningful detection thresholds
- Reducing false positives without missing real threats

14. Future Enhancements

- Real-time alerting with configurable thresholds
- Email notifications for critical security events
- Risk-based alerting using Splunk ES
- Threat Intelligence feed integration
- Geo-location analysis of attacker IP addresses
- Automated incident response workflows
- Splunk Enterprise Security (ES) full integration

15. Conclusion

This project demonstrates the practical use of Splunk SIEM for detecting SSH brute force attacks. By analyzing Linux authentication logs and correlating failed and successful login events, the solution enables SOC analysts to identify suspicious activity and respond to potential security incidents efficiently.

The project provides valuable hands-on experience in:

- Log analysis and normalization
- SIEM operations with Splunk
- Threat detection using SPL queries
- Incident investigation workflows

These competencies are directly applicable to SOC Analyst and Cybersecurity roles, making this project a strong foundation for professional development in the field.

16. References

15. Splunk Documentation — docs.splunk.com
16. MITRE ATT&CK Framework — attack.mitre.org
17. Linux Authentication Logging Documentation
18. NIST Cybersecurity Framework (CSF)
19. OWASP Authentication Security Guidelines